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Homogeneous Catalysis

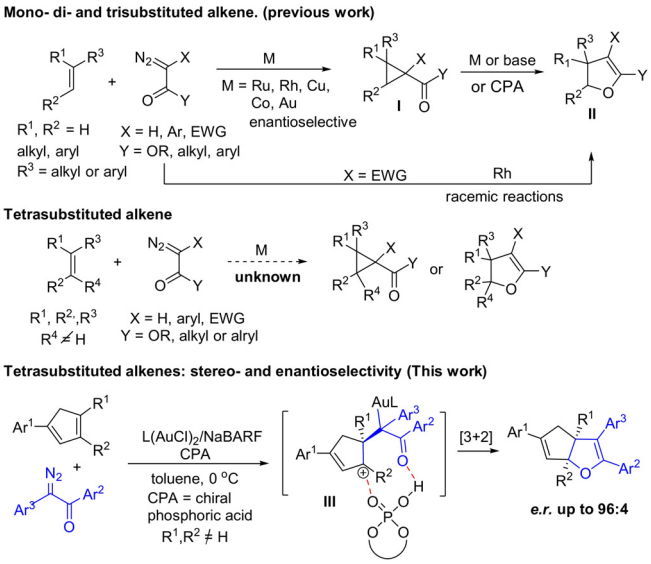
Gold-Catalyzed [3+2]-Annulations of α-Aryl Diazoketones with the Tetrasubstituted Alkenes of Cyclopentadienes: High Stereoselectivity and Enantioselectivity

Ching-Nung Chen, Wei-Min Cheng, Jian-Kai Wang, Tzu-Hsuan Chao, Mu-Jeng Cheng, and Rai-Shung Liu**

Abstract: *This work reports gold-catalyzed [3+2]-annulations of α-diazo ketones with highly substituted cyclopentadienes, affording bicyclic 2,3-dihydrofurans with high regio- and stereoselectivity. The reactions highlights the first success of tetrasubstituted alkenes to undergo [3+2]-annulations with α-diazo carbonyls. The enantioselective annulations are also achieved with high enantioselectivity using chiral forms of gold and phosphoric acid. Our mechanistic analysis supports that cyclopentadienes serve as nucleophiles to attack gold carbenes at the more substituted alkenes, yielding gold enolates that complex with chiral phosphoric acid to enhance the enantioselectivity.*

Metal-catalyzed cyclopropanations of alkenes with α-diazo carbonyls are powerful tools to form cyclopropane derivatives,[1] which serve as versatile building blocks for numerous bioactive compounds.[2] These catalytic cyclopropanations have been performed with excellent enantioselectivity on mono-, di-, and trisubstituted alkenes (Scheme 1).[1–3] One important application[4] of the resulting chiral cyclopropanes (**I**) is a subsequent transformation (Cloke–Wilson rearrangement) into 2,3-dihydrofurans (**II**) using Rh^{II} catalyst or strong bases, but a loss of enantiopurity readily occurs. With chiral phosphoric acid (CPA), enantiopure 2,3-dihydrofurans (II) could be obtained from racemic cyclopropane species (**I**) through kinetic resolutions.[5] Although direct productions of 2,3-dihydrofurans from mono-, di-, or trisubstituted alkenes and special α-diazo carbonyls (X = EWG) have been reported, but are strictly limited to racemic products.[6] The main limitation of current alkene cyclopropanations is the catalytic inefficiency with tetrasubstituted alkenes with a metal catalyst,[7] even for racemic versions. To seek a breakthrough, we report the first success of catalytic [3+2]-annulations of tetrasubstituted alkenes with α-diazo ketones with high efficiency. To enhance the synthetic value, we have developed their enantioselective [3+2]-annulations using dual chiral catalysts, including, gold complexes and phosphoric acids (CPA). A cooperative effect of the two catalysts enables us to postulate a gold enolate intermediate (**III**) bearing also an allylic cation, which rationalizes well the preferred regioselectivity at the more substituted alkenes of cyclopentadienes.

As shown in Figure 1, this hexahydro-2*H*-cyclopenta-[*b*]furan core exists in several sesquiterpenes species **IV 1–5**



Scheme 1. The annulations of α-diazo ketones with tetrasubstituted alkenes.

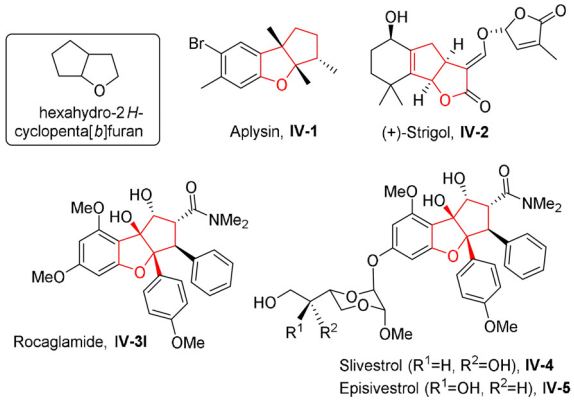


Figure 1. Selected bioactive molecules containing hexahydro-2*H*-cyclopenta[*b*]furan core.

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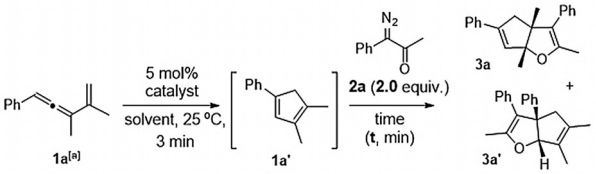
that exhibit biological activity. Among them, aplysin (**IV-1**) protects liver cells against alcohol-induced injury,[8a] (+)-strigol, (**IV-2**) is a potent root-gemeniation stimulant of several weeds.[8b] Rocaglamide (**IV-3**) was found to have a cytostatic effect in T cells infected with HTLV-1 associated T-cell leukemia.[8c] Silvestrol (**IV-4**) and its diastereoisomer Episivestrol (**IV-5**) showed activity in the P-388 lymphocytic leukemia test system in vivo.[8d]

We optimized the [3+2]-annulations of α-diazo methylketone **2a** with vinylallene **1a**, a precursor of substituted cyclopentadiene **1a′**.[9] Vinylallene **1a** was initially treated with a gold catalyst in DCE (5 mol%, 25 °C) for 3 min to ensure a complete formation of cyclopentadiene **1a′** before diazo species **2a** was added to complete a subsequent [3+2]-annulation. With LAuCl/AgOTf (L = P(*t*-Bu)$_2$(*o*-biphenyl), a [3+2]-annulations product **3a** was obtained in 36% (Table 1, entry 1). With less electron-rich LAuCl/AgX (L = $P(OPh)_3$ and PPh_3), compound **3a** was obtained in 37 and 48% yields, respectively. Notably, another regioisomer **3a"** was isolated in 18–24% yields (entries 2–3). IPrAuCl/AgOTf (5 mol%) yielded compound **3a** in 68% with byproduct **3a′** in negligible yields (<2%, entry 4). For other silver salts in IPrAuCl/AgX (X = NTf_2 and SbF_6), the annulation product **3a** was increased to 74–78% yields, with $AgSbF_6$ being the most efficient. IPrAuCl/NaBARF (5 mol%) gave compound **3a** in 71% yield (entry 7). $AgSbF_6$ alone failed to generate cyclopentadiene **1a′** (entry 8). IPrAuCl/$AgSbF_6$ in other solvents afforded our target **3a** with the following yields: DCM (56%), THF (41%) and toluene (29%; entries 9–11). We also tested diphosphinegold catalyst dppp$(AuCl)_2$/$AgSbF_6$ (5 mol%), yielding compound **3a** in 68% and **3a′** in 6% (entry 12). We isolated cyclopentadiene **1a′** that reacted with diazo species **2a** with IPrAuCl/$AgSbF_6$ (5 mol%) in DCE to yield the desire product **3a** in 75% (entry 13). With $Rh_2(OAc)_4$, the reaction of cyclopentadienes **1a′** with diazo ketone **2a** failed to give any noticeable product (entry 14) while $Cu(OTf)_2$ yield a mixture of products **3a** and **3a′** in 16% and 8% yields, respectively (entry 15).[9] The molecular structure of compound **3a** is inferred with X-ray diffraction study of its relatives **5e** and (+)-**6q**.[10]

We assessed the substrate scope using various vinylallenes **1** and α-diazo methylketone **2a** under optimized conditions; the results are shown in Table 2. Herein, all the [3+2]-annulations occurred only at tetrasubstituted alkenes of cyclopentadienes **1′.** Vinylallenes **1b–1f** bearing various *para*-phenyl substituenents 4-XC_6H_4 (X = OMe, Me, Br, Cl and CF_3) afforded products **3b–3f** in 52–67% yields (entries 1–5). These annulations were also operable with 3- and 2-substituted phenyl derivatives **1g–1j** (X = OMe, Me and Cl, *R*1 = 2-MePh), yielding the annulation products **3g–3j** in 62–68% yields (entry 6–9). For vinylallene **1k** bearing *R*1 = 2-naphthyl, its resulting compound **3k** was obtained in 59% (entry 10). We also prepared 3-thienyl derivative **1l** that afforded compound **3l** in 62% yield (entry 11). With R^1 = phenyl, we varied R^2 or R^3 of vinylallenes **1m–1n** with alkyl groups, their resulting product **3m–3n** were produced with 42–48% (entries 12, 13). We also tested the reactions with all-alkyl substituted vinylallenes **1o–1p** (R^1 = *i*-Pr and Cy), further produced the desired products **3o–3p** in 46–49 yields (entries 14, 15).

We further assessed the reaction generality with various α-diazo ketones **2** (Table 3). Again, only tetrasubstituted alkenes of cyclopentadienes are the reactive sites. For α-diazo methylketones **2b–2e** (*R*1 = 4-XPh, X = Me, Br, Cl and F),

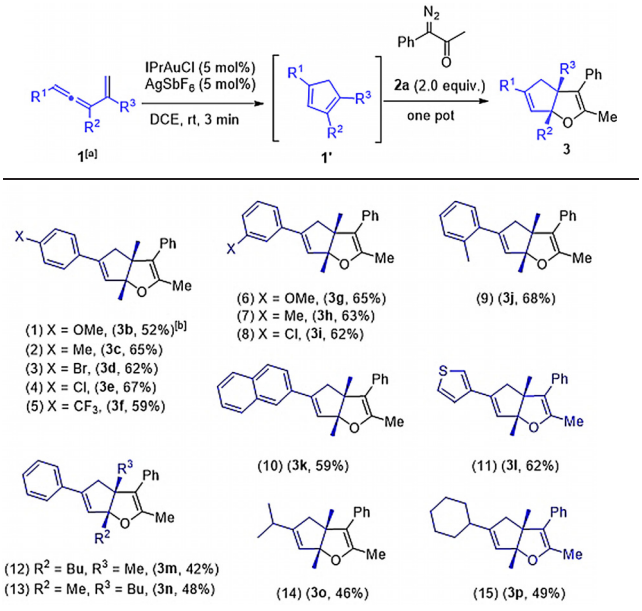
Table 1: One-pot catalytic reactions with various metal catalysts.



Entry	Catalyst	*t* [min]	Solvent	Yields [%][b] 1a′	3a	3a′
1	LAuCl/AgOTf	20	DCE	28	36	–
2	$(PhO)_3PAuCl$/AgOTf	20	DCE	–	37	18
3	PPh_3AuCl/AgOTf	30	DCE	–	48	24
4	IPrAuCl/AgOTf	10	DCE	5	68	–
5	IPrAuCl/$AgNTf_2$	10	DCE	–	74	–
6	IPrAuCl/$AgSbF_6$	10	DCE	–	78	–
7	IPrAuCl/NaBARF	60	DCE	–	71	–
8	$AgSbF_6$	60	DCE	0	–	–
9	IPrAuCl/$AgSbF_6$	15	DCM	–	56	–
10	IPrAuCl/$AgSbF_6$	20	THF	27	41	–
11	IPrAuCl/$AgSbF_6$	30	Toluene	35	29	–
12	dppp$(AuCl)_2$/$AgSbF_6$	60	DCE	–	68	6
13	IPrAuCl/$AgSbF_6$[c]	20	DCE	–	75	–
14	$Rh_2(OAc)_4$[c]	30	DCE	36	–	–
15	$Cu(OTf)_2$[c]	180	DCE	5	16	8

[a] [**1a**] = 0.15 M. [b] Product yields are obtained after purification from a silica column. [c] Entries 13–15 refer to the reactions using cyclopentadi-enes **1a′** as the reactant. L = P(*t*-Bu)2(*o*-biphenyl). IPr = 1,3-bis(diisopro-pylphenyl)imidazol-2-ylidene). Tf = trifluoromethanesulfonyl. dppp = 1,3-Bis(diphenylphosphino)propane.

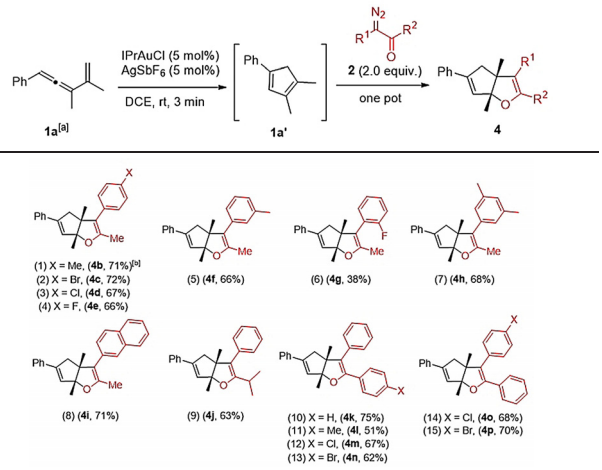
Table 2: [3+2]-annulations with various vinyl allenes.



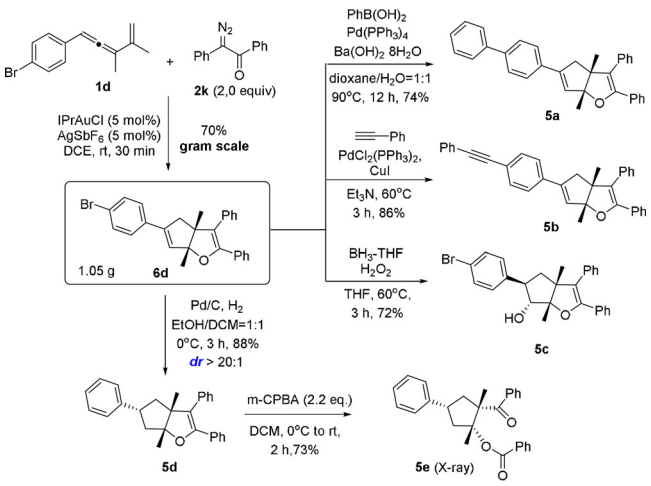
[a] [**1**] = 0.15 M. [b] Product yields are obtained after purification from a silica column. IPr = 1,3-bis(diisopropylphenyl)imidazol-2-ylidene).

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Table 3: Catalytic [3+2] annulations with various α-diazo ketones.



[a] [**1a**] = 0.15 M. [b] Product yields are obtained after purification from a silica column. IPr = 1,3-bis(diisopropylphenyl)imidazol-2-ylidene).



Scheme 2. Gram-scale synthesis and chemical functionalization.

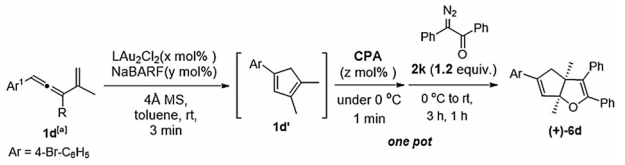
their resulting compound **4b–4e** were obtained in 66–72% yields (entries 1–4). The annulations worked well with *meta*-methyl and *ortho*-fluoro phenyl derivatives **2f–2g**, forming compounds **4f** and **4g** in 66% and 38% yields, respectively (entries 5, 6). For 3,5-dimethylphenyl derivative **2h**, its corresponding product **4h** was obtained in 68% yield (entry 7). We tested the reaction with 2- naphthylalkynyl substrate **2i** that afforded compound **4i** in 71% yield (entry 8). The reactions were further amenable to α-diazo isopropylketone **2j**, yielding compound **4j** in 63% yield (entry 9).

For α-diazo phenylketone **2k**, its resulting compound **4k** was produced in 75% yield (entry 10). For *para*-phenyl ketones (R^2 = 4-XPh, X = Me, Cl and Br) of diazo species **2l–2n**, their resulting bicyclic furans **4l–4n** were obtained in 51–67% yields (entries 11–13). For α-phenyl diazoketones **2o–2p** bearing *para*-phenyl derivatives (X = Cl, Br), their annulation products **4o–4p** were obtained in satisfactory yields (68–70%, entries 14, 15). Other α-phenyl diazo ester $PhC(N_2)CO_2Et$ was an inapplicable substrate to give three inseparable products.

Scheme 2 shows a gram-scale synthesis and chemical functionalization of one representative **6d**. Compound **6d** was prepared on 1.05 g with a yield up to 70%. The Suzuki reaction of **6d** yielded the phenyl coupling product **5a** in 74%. Alternatively, the Sonogashira reaction proceeded smoothly with species **6d** to afford compound **5b** in 86% yield. Hydroboration of species **6d**, followed by H_2O_2 oxidation delivered an alcohol **5c** stereoselectively (dr > 20:1). Selective hydrogenation of compound **6d** in EtOH/DCM at 0 °C led to formation of compound **5d** with dr > 20:1. Oxidative cleavage of compound **5d** was achieved with *m*-CPBA (2.2 equiv) in DCM, generating a functionalized cyclopentane **5e** in 73% yield. The structure of compound **5e** was confirmed with X-ray differaction.[10]

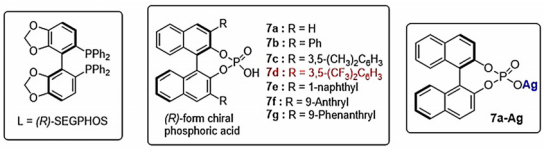
To realize the [3+2]-enantioselective annulations, we first screened seven chiral phosphine gold catalysts (see the Supporting Information) and found that L(AuCl)$_2$/NaBARF (5.5 mol%, L = (*R*)-SEGPHOS) was the most effective to induce the enantioselectivity, rendering (+)-**6d** with 78% yield and e.r. = 78:22 in cold toluene (0 °C, 3 h; Table 4, entry 1). Notably, this e.r. ratio was increased to 87:13 in the presence of 5% chiral phosphoric acid **7a** (entry 2). We noticed no change of enantioselectivity when gold catalyst, NaBARF and CPA **7a** were used with 2.0–2.2 mol%. (entry 3). An increase of CPA **7a** alone (4.0 mol%) improved the e.r. = 88:12 (entry 4). We proceeded through the family of BINOL-based CPAs **7b–7g** (entries 5–10), observing that

Table 4: One-pot chiral catalytic reactions with various CPA catalysts.



Entry	L-$(AuCl)_2$ (x mol%)	NaBARF (y mol%)	CPA (z mol%)	Yields[b] [%]	*er* ratio[c] (+)/(−)
1	5.5	5.0	none	78	78:22
2	5.5	5.0	**7a** (5.0)	75	87:13
3	2.2	2.0	**7a** (2.0)	72	87:13
4	2.2	2.0	**7a** (4.0)	70	88:12
5	2.2	2.0	**7b** (4.0)	70	88:12
6	2.2	2.0	**7c** (4.0)	74	91:9
7	2.2	2.0	**7d** (4.0)	73	93:7
8	2.2	2.0	**7e** (4.0)	75	89:11
9	2.2	2.0	**7f** (4.0)	72	89:11
10	2.2	2.0	**7g** (4.0)	68	87:13
11[d]	2.2	–	**7a-Ag** (2.0)	21	82:18

[a] [**1d**] = 0.125 M. [b] Product yields are obtained after purification from a silica column. [c] Determined by HPLC analysis on a chiral stationary phase (see the Supporting Information). [d] Cyclopentadiene **1d'** is used as starting material. L = (*R*)-SEGPHOS.

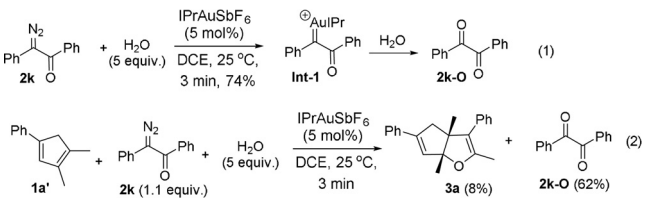


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a chiral gold catalyst together with CPA **7d** (4 mol%, R = 3,5-$(CF_3)_2$Ph) yielded desired (+)-**6d** with e.r. = 93:7 (entry 7). A decrease in reaction temperature or an increased loading of chiral acid **7d** (6 mol%) led to decreased yields of species (+)-**6d**. In entry 11, chiral phosphate **7a-Ag** replaced NaBARF, but compound (+)-**6d** was obtained in only 21% yield with e.r. = 82:18 (entry 11). Brønsted acid is indispensable to obtain high product yields. The effects of silver salts and other solvents were also studied with the results in the Supporting Information, Tables S2, S3. The absolute configuration of our target (+)-**6d** is inferred from its relative (+)-**6q** (see below).

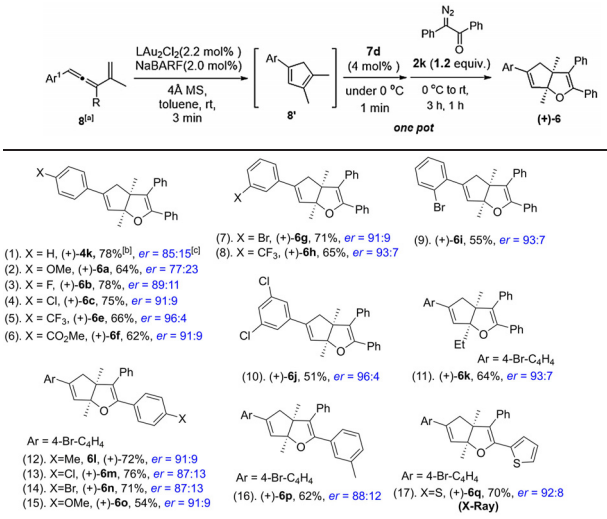
With these optimized condition, such enantioselective annulations were conducted with various vinylallenes **8** and α-diazo aryl ketones **2**; the results are presented in Table 5. We first tested the reaction with vinylallene **1a** and α-diazo phenyl ketone **2k**, yielding species (+)-**4k** in 78% yield with e.r. = 85:15. An electron-rich *para*-methoxyphenyl containing vinylallene **1b** delivered species (+)-**6a** with a low e.r. = 77:23. In contrast, electron-deficient phenyl analogues **8b**, **1e**–**1f** and **8f** (4-XC_6H_4, X = F, Cl, CF_3 and CO_2Me) afforded desired products (+)-**6b**–**6f** in satisfactory yields (ca. 62–78%) and high e.r. = 89:11–96:4 (entries 3–6). For *meta*-Br, CF_3 and *ortho*-Br substituted phenyl-vinylallenes **8g**–**8i**, their resulting products (+)-**6g**–**6i** were obtained with e.r. = 91:9–93:7 (entries 7–9). 3,5-Dichlorophenyl-derived vinylallene **8j** was very suitable to this annulation, giving compound (+)-**6j** with e.r. = 96:4 (entry 10). With Ar^1 = 4-BrPh, we varied the C(4)-R substituent of vinylallene with an ethyl group, affording the desired products (+)-**6k** in 64% yield with e.r. = 93:7 (entry 11). These enantioselective annulations were compatible with various α-diazo phenylketones **2l**–**2n** and **2q**–**2r**, bearing aryl = 4-XC_6H_4 (X = Me, Cl, Br, and OMe) and 3-MeC_6H_4, affording the desired products (+)-**6l**–**6p** in satisfactory yields 54–72% with e.r. = 87:13–91:9 (entries 12–16). For 2-thienylketone **2s**, its corresponding product (+)-**6q** was produced with e.r. = 92:8 (entry 17). The absolute configuration of compound (+)-**6q** was characterized with X-ray diffraction.[10]

We performed control experiments Equations (1), (2) to support the intermediacy of gold carbenes **Int-1**, Treatment of α-diazo ketone **2k** with water gave 1,2-diphenyldiketone **2k-O** as the main product [Eq. (1)]. A rapid formation of gold carbenes **Int-1** from α-diazo ketone **2k** was evident, which was subsequently oxidized with water to give diketone **2k-O** [Eq. (1)]. In our [3+2]-annulation reaction, the presence of water also led to formation of this diketone species in 62% yield, indicative of gold carbene intermediates [Eq. (2)].

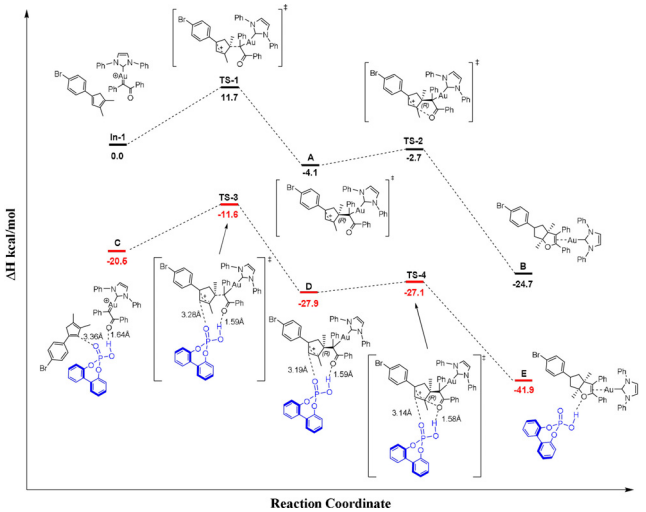


With DFT calculations, we postulate a new mechanism in Scheme 3 involving a nucleophilic attack of cyclopentadienes 1′ at gold carbenes **In-1** to generate gold-enolates **A** comprising an allyl cation. In this route, the regioselectivity at the more substituted alkene is preferable because the resulting allyl cation is greatly stabilized by 1-methyl and 3-aryl substituents. A subsequent ring closure of species **B** will form the annulation intermediate **A**. Our DFT calculations with simplified ligands and acids in toluene indicate an exothermic process for this annulation, **In-1**→**A**→**B**, with two activation barriers ca. 11.7 kcal mol^{-1} and 1.4 kcal mol^{-1}, respectively. In the presence of phosphoric acid, gold carbenes **In-1** and cyclopentadienes **1′** can react with this acid to form a complex pair **C**; this process releases 20.5 kcal mol^{-1}. Such a acid-bound species **C** is very favorable to form gold enolate **D** that is stabilized with this phosphoric acid, as manifested by its low energy state (−27.9 kcal mol^{-1})

Table 5: Enantioselective [3+2]-annulations.



[a] [**8**] = 0.125 M. [b] Product yields are obtained after purification from a silica column. [c] Determined by HPLC analysis on a chiral stationary phase (see the Supporting Information). L = (*R*)-SEGPHOS.



Scheme 3. DFT calculations on catalytic [3+2]-annulations.

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and a small kinetic barrier (+ 8.9 kcal mol^{-1}). In the optimized geometry of intermediate **D**, the hydroxyl of phosphoric acid is bound to the ketone of enolates **D** and its basic P=O oxygen stabilizes the allylic cation. A final ring closure of species **E** has an activation energy, ca. only 0.8 kcal because of a preorganized geometry. Accordingly, the presence of phosphoric acid facilitates this [3+2]-annulation via formation of ion pairs.

This mechanism also rationalizes well the high enantiopurity of compounds **(+)-6** with cyclopentadienes **1′** bearing electron-deficient aryl groups (Ar = X-C_6H_4, X = Cl, Br, and CF_3) because their allylic ***C*-Me** ends become highly positive to form a tightly bound pair with the basic P=O of phosphate. We also performed DFT calculation involving a prior dissociation of IPrAu$^+$ from intermediates **D**; the energy profiles of these pathways are unlikely to occur (Supporting Information, Figures S3, S4).

Before this work, tetrasubstituted alkenes are typically inactive toward common α-diazo carbonyl species in either cyclopropanations or [3+2]-annulations. Herein, we report gold-catalyzed [3+2]-annulations of α-diazo ketones with substituted cyclopentadienes;[11] the regioselectivity occurs exclusively on the tetrasubstituted alkene ends. To light the synthetic value, we achieved high enantioselective[12] [3+2]-annulations using a combination of chiral gold catalysts and chiral phosphoric acids. We postulate a mechanism involving[13,14] a nucleophilic attack of cyclopentadienes at gold carbenes, generating gold enolates bearing allylic cations, which can complex with chiral phosphoric acid to increase the enantioselectivity.

Acknowledgements

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Conflict of interest

The authors declare no conflict of interest.

Keywords: [3+2]-annulations · Cloke–Wilson rearrangement · gold catalysis · tetrasubstituted alkenes · α-diazo ketones

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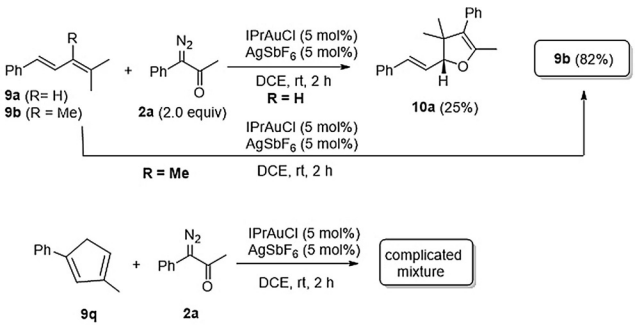
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[14] Additional data to manifest the effects of solvents, phosphoric acids, and silver salts on asymmetric [3+2]-annulations are provided in the Supporting Information, Tables S2, S3 and Eqs. (s1),(s2).

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